

Predicting Life-Course Trajectories from Early Adult Histories and Family Backgrounds

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Abstract—We predict adult life-course trajectories in earnings, employment, and marriage from each individual’s first decade of adult observations and their family-of-origin background, using the longitudinal data of Panel Study of Income Dynamics (PSID) 1960–1975 cohort. Rather than predicting a single endpoint, we predict a coupled sequence of outcomes across ages 30–45, framed as a multi-task sequence-to-sequence problem. Our main model is a Task-Decoder Transformer in which a shared self-attention encoder summarizes the input history, and three task-specific decoders use cross-attention to query that history alongside static demographic and parental tokens. We compare three decoding strategies—parallel (one shot), autoregressive (feed generated outcomes forward), and scheduled-sampling autoregressive—against the mean/majority baseline. On the held-out test set, the parallel task-decoder is the most stable variant; the autoregressive variants underperform it, and scheduled sampling does not recover the gap. An input ablation shows that early-adult history alone recovers most of the predictive signal in earnings ($R^2 \approx 0.337$ vs. 0.367 with full inputs), while static parental context alone collapses to $R^2 \approx 0.144$, supporting a near-Markov reading in which family background acts mainly through early trajectories. Across all models, predictive accuracy declines monotonically with the prediction horizon.

 **GitHub Repository:** https://github.com/ZixiLi76/CSCI1470_FinalProject_TokensToRiches

I. INTRODUCTION

The transition from early to mid adulthood is the period when life trajectories begin to diverge: some individuals build stable careers, accumulate wealth, and form lasting families, while others experience labor-market detachment, instability, or shocks whose consequences persist for decades. Quantifying how predictable these adult trajectories are from a person’s first decade of adult observations, and how much of that predictability is determined upstream by family of origin, speaks directly to questions about intergenerational mobility, the timing of inequality formation, and the structure of life-course processes.

A. Motivation

Most existing prediction work in this area treats outcomes as single endpoints (e.g., earnings at age 40, marital status at age 45) and uses cross-sectional or short-panel regression. Our motivation is twofold. First, life-course outcomes are coupled: a person’s earnings, employment, and family status at age 35 are not independent. Second, adulthood is

a sequence, not a snapshot. We wanted to know whether the same neural sequence-modeling tools that have proven effective for natural language and tabular time-series can recover meaningful structure from sparse, irregularly observed life histories, and whether predicting full trajectories rather than endpoints reveals where signal lives across the life course.

B. Goals

- **Base Goal:** Build an end-to-end pipeline from raw PSID-SHELF microdata to a working Transformer that predicts at least one outcome (earnings) at one age (45) better than a flat-features linear baseline.
- **Target Goal:** Train a multi-task Transformer that jointly predicts earnings, employment, and marital status across multiple future age bins, with rigorous comparisons to mean/majority baseline using identical features.
- **Stretch Goal:** Compare multiple decoding strategies (parallel, autoregressive, scheduled-sampling) to test whether life-course prediction benefits from autoregressive generation in the way natural language does.

We met all three goals; the stretch comparison ended up producing the most interesting (and partly negative) finding of the project.

C. Contributions

This paper contributes (1) a Task-Decoder Transformer architecture for multi-task life-course trajectory prediction with shared encoder and outcome-specific cross-attention decoders, (2) a controlled comparison of three decoding strategies on the same backbone, (3) an input-ablation analysis showing that early-adult history mediates most of the family-background signal, and (4) a fully reproducible PSID-SHELF preprocessing pipeline released alongside the code.

II. LITERATURE REVIEW

Sequence models have a long history in NLP [1], [2] and increasingly in panel and longitudinal applications outside language. Pang *et al.* [7] use recurrent networks on electronic health records to predict downstream clinical outcomes; Savcisen *et al.* [6] apply Transformer-style architectures to large-scale Danish administrative panel data to forecast life events including mortality and earnings. These works share our framing of life events as sequential tokens but differ in scale and target structure.

In economics and demography, sequence-analysis methods (Optimal Matching, multichannel sequence analysis) have been used descriptively for decades [8], [9], but they are typological rather than predictive. Predictive work on PSID has historically used Mincer regressions [10] or hierarchical linear models for earnings dynamics; deep learning applications to PSID remain rare.

On the architectural side, the choice between RNNs, LSTMs, and Transformers for sequential prediction has been studied extensively. Recurrent gated cells (LSTM [3]) address vanishing gradients in vanilla RNNs but process input sequentially; self-attention [1] processes the sequence in parallel and can directly attend to any input timestep, which is attractive when "key moments" (e.g., a job loss or a marital event) carry disproportionate predictive weight. We use Transformers for this reason. Scheduled sampling [4] was proposed to mitigate exposure bias in autoregressive sequence generation, motivating one of our decoding variants.

Our project differs from prior work in three ways: (1) we predict *linked multi-task trajectories* rather than a single endpoint or single-outcome sequence; (2) we use *cross-attention* between task-specific decoders and a shared encoder, allowing outcomes to share upstream representation while keeping outcome-specific prediction rules; and (3) we directly contrast decoding strategies on the same backbone, isolating the effect of generation strategy from architecture.

III. METHODOLOGY

A. Dataset

We use the Panel Study of Income Dynamics (PSID) Social, Health, and Economic Longitudinal File (PSID-SHELF, v2025-01) [11], which harmonizes 42 waves of the PSID from 1968 to 2021. We restrict the sample to individuals born 1960–1975 who were ever observed as a member of the PSID reference couple (reference person or spouse/partner; REFCOUPLE=1) because we would have complete for this group of people from ages 20 to 45. After applying coverage filters (≥ 8 waves of input observation in ages 20–29 and ≥ 2 waves of target observation in ages 30–45), the analytic sample contains 2,585 unique individuals across 38,286 person-year rows. We split individuals 8:1:1 (train/val/test) with a fixed random seed; all preprocessing statistics (means, standard deviations, category vocabularies, percentile cutoffs) are fit on training data only to prevent leakage.

PSID has annual waves through 1996 and biennial waves thereafter, so observation intervals are not uniform. To harmonize these we bin observations into five two-year input bins, [20–21], [22–23], [24–25], [26–27], [28–29], and eight two-year target bins, [30–31], [32–33], ..., [44–45]. Each age bin is one vector. Where multiple waves fall in a single bin, we average; where none do, the bin is marked missing.

B. Features and Preprocessing

For each individual we construct three input groups:

- **History (sequence) features over ages 20–29:** log-earnings, employment status (categorical), marital status (categorical), and the binary observation mask for each.
- **Static parental context:** parents' mean income and income volatility during the ego's ages 10–18, parents' highest education, parents' marital trajectory (share married, share divorced), and observation counts.
- **Static demographic context:** sex, race/ethnicity, region of childhood, birth-cohort bin.

Earnings are transformed onto the signed-log scale, $\tilde{y} = \text{sign}(y) \log(1 + |y|/1000)$, and then standardized using the training mean and standard deviation. Continuous parental and demographic variables are standardized similarly. Categorical variables are mapped to integers using vocabularies built from the training set, with reserved codes 0 (missing) and 1 (unseen-at-test-time).

C. Model Architecture

Shared backbone. Let $\mathbf{h}^{\text{hist}} \in \mathbb{R}^{T_{\text{in}} \times d}$ denote the encoded history with $T_{\text{in}} = 5$ input bins and embedding dimension d , and let $\mathbf{h}^{\text{stat}} \in \mathbb{R}^{S \times d}$ denote the encoded static context with one token per static feature. The history encoder fuses continuous and embedded categorical history features per bin, adds sinusoidal positional encodings, and passes the result through a Transformer encoder layer:

$$\mathbf{h}^{\text{hist}} = \text{TransformerEncoder}(\text{PE}(\mathbf{W}^{\text{hist}} \mathbf{x}^{\text{hist}})), \quad (1)$$

where $\mathbf{W}^{\text{hist}}$ is a linear projection from raw bin features to d dimensions. Self-attention within the encoder allows the model to weight, for example, an unstable employment record at age 23 against stable earnings at age 28, with weights that depend on both content and position.

The static context is tokenized similarly: each continuous feature contributes one token (value-modulated by a learned feature embedding) and each categorical feature contributes one token (categorical embedding projected to d). This converts heterogeneous tabular features into a uniform token bag the decoder can attend over.

Task-specific decoders. For each outcome $o \in \{\text{earnings, employment, marriage}\}$, the decoder maintains $T_{\text{out}} = 8$ learned query vectors $\mathbf{Q}_o \in \mathbb{R}^{T_{\text{out}} \times d}$, one per future bin. Each decoder block applies cross-attention to the history memory, then cross-attention to the static memory, then a position-wise feedforward network with residual connections and layer normalization:

$$\mathbf{Q}_o \leftarrow \text{LN}(\mathbf{Q}_o + \text{Attn}(\mathbf{Q}_o, \mathbf{h}^{\text{hist}}, \mathbf{h}^{\text{hist}})), \quad (2)$$

$$\mathbf{Q}_o \leftarrow \text{LN}(\mathbf{Q}_o + \text{Attn}(\mathbf{Q}_o, \mathbf{h}^{\text{stat}}, \mathbf{h}^{\text{stat}})), \quad (3)$$

$$\mathbf{Q}_o \leftarrow \text{LN}(\mathbf{Q}_o + \text{FFN}(\mathbf{Q}_o)). \quad (4)$$

The output of each decoder is passed to an outcome-specific head: a linear projection to a scalar for earnings (with MSE loss), and a linear projection to class logits for employment and marriage (with cross-entropy loss).

Decoding variants. The shared encoder is identical across variants; only the decoder differs:

- **Parallel:** decoder predicts all T_{out} bins in one forward pass with no cross-bin dependencies; the only signal flowing across target ages is via the shared encoder.
- **Autoregressive:** decoder predicts bin $t + 1$ conditional on its own predictions for bins $1, \dots, t$, fed back as additional decoder input through a self-attention mask.
- **Scheduled-sampling autoregressive:** same as autoregressive, but during training the decoder probabilistically substitutes its own predictions for the ground-truth previous bins with probability that increases linearly across epochs.

D. Training Procedure

The multi-task loss combines a masked MSE for earnings with masked cross-entropy for employment and marriage:

$$\mathcal{L} = w_e \cdot \text{MSE}^m(\hat{y}_e, y_e) + w_p \cdot \text{CE}^m(\hat{y}_p, y_p) + w_m \cdot \text{CE}^m(\hat{y}_m, y_m), \quad (5)$$

with $(w_e, w_p, w_m) = (1.0, 0.5, 0.5)$, selected experimentally as the best-performing combination among several weightings we tried, and the superscript m denoting masked aggregation that ignores missing target bins. We optimize with AdamW [5] and a cosine learning-rate schedule over 60 epochs, with early stopping on the validation earnings MSE (patience 7). Gradient norms are clipped at 1.0. Hyperparameters are tuned by grid search over $d \in \{64, 128\}$, dropout $\in \{0.10, 0.25\}$, learning rate $\in \{3 \times 10^{-4}, 1 \times 10^{-3}\}$, weight decay $\in \{10^{-4}, 10^{-3}\}$ with $n_{\text{heads}} = 4$, one history-encoder layer, and one task-decoder layer held fixed throughout. The best configuration is selected on validation MSE alone; test metrics are reported only after model selection is locked.

E. Baselines

To isolate the gain from sequence modeling, we evaluate mean-majority baseline on identical features:

- **Mean / Majority:** Predict the training-set marginal mean for earnings and the training-set modal class for employment and marriage at each target bin. This is the floor any non-trivial model must clear.

F. Evaluation Metrics

For earnings we report the test-set coefficient of determination R^2 and root-mean-squared error (on the signed-log scale) per target bin. For employment and marriage we report classification accuracy per target bin. Sequence-level metrics are computed by averaging over bins for which the target is observed; missing targets are excluded from both training loss and evaluation metrics.

IV. RESULTS

A. Quantitative Results

Table I summarizes test-set metrics averaged across all target bins. The parallel task-decoder Transformer is the strongest variant on earnings ($R^2 = 0.367$), beating both autoregressive variants. All Transformer variants decisively beat the mean/majority baseline (see Fig. 1).

TABLE I: Test-set metrics averaged across target bins. Earnings R^2 is on the signed-log scale; Employment and Marriage are top-1 accuracy. Bold marks the best value per column. The parallel task-decoder dominates on earnings and marriage; on employment all four models cluster within 0.01 of each other.

Model	Earn R^2	Emp Acc.	Mar Acc.
Mean / Majority	-0.003	0.817	0.644
Scheduled sampling	0.258	0.820	0.740
Autoregressive	0.305	0.826	0.730
Parallel decoder	0.367	0.825	0.764

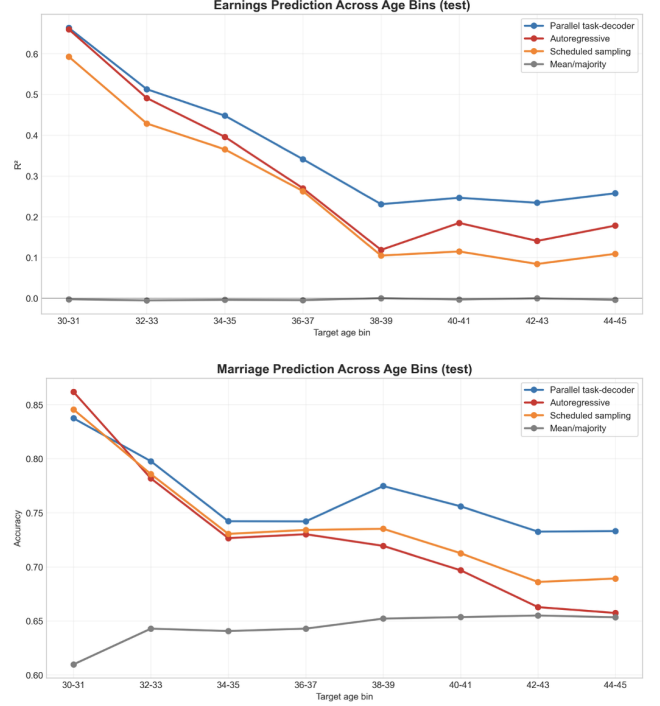


Fig. 1: Test-set R^2 for earnings (top) and accuracy for marriage (bottom) by target age bin. Predictive performance declines monotonically with prediction horizon for all models. The parallel task-decoder degrades most gracefully; the autoregressive variants degrade fastest as generated past predictions accumulate error.

B. Input Ablation

To understand where the predictive signal lives, we retrain the parallel task-decoder under three input regimes (Fig. 2). With the full input (history + static context), test earnings R^2 is 0.367. Dropping the static parental and demographic context but keeping the early-adult history barely hurts: R^2 falls only to 0.337. In contrast, dropping the history but keeping static context collapses performance to $R^2 = 0.144$. Family background therefore explains substantially less unique variance than the early career itself, and most of the family signal appears to be already encoded in early adulthood.

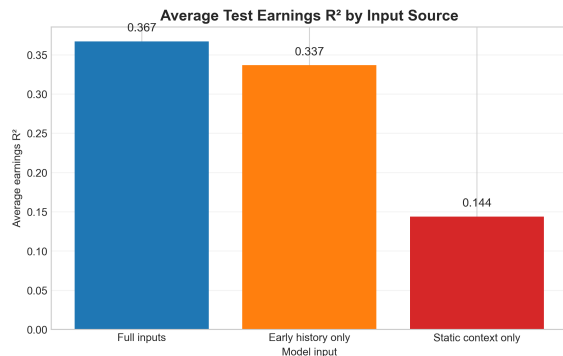


Fig. 2: Input-ablation test-set earnings R^2 . Early-adult history alone recovers most of the full-input performance; static context alone is much weaker. This is consistent with a near-Markov reading in which family background influences early trajectories which then mediate later outcomes.

C. Qualitative Analysis

Three qualitative patterns are consistent across runs. First, predictions are sharper for earlier target bins and degrade with horizon—an expected consequence of accumulating uncertainty. Second, the parallel decoder degrades more gracefully than the autoregressive variants: when the autoregressive decoder’s prediction at bin t is wrong, that error is fed in as input at bin $t + 1$ and amplifies. Third, scheduled sampling does not close the gap. We had hypothesized that exposing the model to its own predictions during training would improve robustness at inference, as it does in language modeling, but in this data setting the additional training noise outweighs the exposure-bias correction.

D. Goal Achievement

We met all three goals. The base goal (a Transformer beating linear baselines on at least one outcome at one age) was met early in development. The target goal (multi-task multi-bin prediction with disciplined linear baselines) is reflected in Table I and Fig. 1. The stretch goal (decoding-strategy comparison) is the project’s most interesting result: the negative finding that autoregressive decoding does not help is unexpected and substantively meaningful for life-course modeling.

V. CHALLENGES

Data sparsity and missingness. PSID’s mixed annual/biennial cadence and respondent attrition create irregular observation patterns: a typical individual has 12–14 of a possible 20 person-year rows in the input window, with no two individuals having identical observation patterns. We handled this with two-year binning, per-bin observation masks, and loss masking, but the resulting effective sample is small enough ($N \approx 2,000$ in train) that variance across random seeds is non-negligible.

Parental coverage. Parental features are missing for $\approx 34\%$ of egos because their parents exited PSID before the ego’s ages

10–18 window or were never sample persons. We retain these egos with a `PAR_BOTH_MISS` indicator and zero-impute the parental columns; the ablation analysis confirms the trade-off was acceptable, since the parental block adds only marginal R^2 on top of own history.

Multi-task loss balancing. The earnings (regression), employment (2-class), and marriage (6-class) losses live on different scales. We selected loss weights (1.0, 0.5, 0.5) by validating that earnings did not dominate gradient direction, but a more principled multi-objective scheme (*e.g.*, GradNorm) would likely improve the categorical heads further.

Autoregressive instability. Early autoregressive runs were unstable across seeds: some converged near the parallel decoder’s level, others regressed to predicting the training mean for late bins. Gradient clipping and a more aggressive learning-rate schedule helped, but the variance never matched the parallel decoder’s. This stability gap is itself the headline finding.

VI. ETHICS

Predictive models that link family background to adult outcomes carry real risks. Used in selection contexts such as underwriting, college admissions, hiring, a model that predicts future earnings from parental income would systematically disadvantage individuals whose parents had lower observed earnings, even when the prediction is statistically accurate. Our input-ablation result (Fig. 2) shows that parental context alone has weak predictive power once early-adult history is in the model, but in deployment one cannot assume such history is observed; a partial-information model could lean disproportionately on parental features.

For these reasons we treat parental income, parental education, and parental marital trajectory as measured proxies for structural advantage rather than as features describing individual potential, and we do not advocate any deployment of this model in selection contexts. We also do not release trained model weights publicly, only code and a reproducible preprocessing recipe; PSID microdata itself is restricted-use and cannot be redistributed.

The PSID is collected under informed-consent protocols, and all data we use is de-identified at the participant level. Computationally the project is modest, as the final Transformer trains in under ten minutes on a single Colab T4 GPU, so the carbon footprint of model selection and ablations is negligible. The most consequential ethical choice we made was the framing one: by treating life-course outcomes as predictable from early signals, we risk encouraging a fatalistic reading. We emphasize in the discussion that prediction is not destiny, and that the model’s residual variance ($1 - R^2 \approx 0.6$ on earnings) is itself substantively meaningful and reflects the real role of unobserved shocks, opportunities, and individual choice.

VII. REFLECTION

A. Overall Assessment

The project came together more cleanly than we expected. The biggest surprise was the negative result on autoregressive

decoding: we entered the project assuming that life-course prediction would naturally benefit from the same autoregressive generation that powers language models, and were initially worried when our autoregressive runs underperformed. Once we understood that intermediate-state generation was amplifying noise rather than carrying signal, the result became the most substantively interesting part of the paper.

B. Evolution and Pivots

Our project changed direction twice. First, the original proposal compared LSTM and Transformer encoders against OLS and Random Forest, with five outcomes (earnings, wealth, marital status, family size, occupation) and three prediction modes (single, multi, sequence). We dropped the LSTM and Random Forest baselines as the comparison set grew; we dropped wealth and family size as targets because the multi-task loss was already crowded; and we narrowed the input window from ages 20–40 to 20–29 to make the prediction horizon (ages 30–45) longer and more interesting. Second, we shifted from comparing different model classes (LSTM vs. Transformer) to comparing different decoding strategies on the same backbone, which gave a cleaner controlled comparison.

If we did this over, we would invest more time earlier in the autoregressive variant: by the time we understood why scheduled sampling failed, there was insufficient time to try alternative approaches such as denoising autoregression or diffusion-style generation.

C. Future Work

The most natural extensions are: (1) applying the same architecture to a larger administrative dataset with denser observations and additional outcomes (*e.g.*, health, occupation, geography); (2) experimenting with decoder pretraining on a larger “life-course corpus” before fine-tuning on PSID; (3) replacing the per-bin self-attention with continuous-time attention to handle PSID’s irregular timing without coarse binning; (4) explicitly modeling unobserved shocks as a stochastic component, which our deterministic decoders cannot represent; and (5) continuing the hyperparameter search beyond our current grid, including loss weights, embedding dimension, decoder depth, and longer training schedules, since our compute budget on Colab T4 allowed only a coarse sweep.

D. Key Takeaways

We learned, concretely, that high-dimensional flattened linear baselines are surprisingly competitive on this kind of problem when sample sizes are small. We learned that controlled architectural ablations (same encoder, different decoders) are more informative than headline model-vs-model comparisons. We learned that a rigorous data pipeline, like preregistered hypotheses, train-only preprocessing fits, fixed splits, masked metrics, is what makes the difference between a result one can trust and a result that quietly leaks. And finally, the most humbling takeaway: we cannot finish a person’s life the way a Transformer finishes a sentence, at least not with the data we have. Unlike language, where surrounding tokens reliably

constrain the next, life trajectories carry too much unobserved noise, such as health shocks, macroeconomic conditions, individual choices outside the data, for autoregressive generation to compound information faithfully.

VIII. CONCLUSION

We presented a Task-Decoder Transformer for multi-task life-course trajectory prediction on PSID, and used it to compare three decoding strategies: parallel, autoregressive, and scheduled-sampling, against linear baselines. The parallel decoder is the most accurate and most stable variant; autoregressive generation, even with scheduled sampling, does not improve long-horizon prediction in this setting. An input ablation supports a near-Markov reading of life-course dynamics: family background’s predictive power is largely mediated by early-adult outcomes. Our results are constrained by sample size and missingness, but the methodological pattern: shared encoder, task-specific cross-attention decoders, controlled decoder ablations, transfers to any longitudinal multi-outcome prediction problem.

ACKNOWLEDGMENTS

We thank our mentor TA Hongyang Du for guidance throughout the semester, and the CSCI 1470 teaching staff for helpful feedback at check-in meetings. We are grateful to the Survey Research Center at the University of Michigan for maintaining the PSID and to Daumler, Friedman, and Pfeffer for the harmonized PSID-SHELF release.

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